

Untitled

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Q1

```
con = c(0.0011,0.0014,0.0018,0.0022,0.0010,0.0016,0.0028,0.0020,0.0015,0.0014,0.0023,0.0017,0.0020)
per = c(0.0011,0.0010,0.0019,0.0013,0.0011,0.0017,0.0024,0.0020,0.0013,0.0013,0.0017,0.0015,0.0013)
difference = con - per
t.test(difference, alternative = "greater")
```

```
##
## One Sample t-test
##
## data: difference
## t = 2.6838, df = 12, p-value = 0.009948
## alternative hypothesis: true mean is greater than 0
## 95 percent confidence interval:
## 8.268563e-05 Inf
## sample estimates:
## mean of x
## 0.0002461538
```

by t test, p-value is small, we reject H_0 ## Q2

```
set.seed(123)
obs = c(0.43, 1.17, 0.37, 0.47, 0.68, 0.58, 0.50, 2.75, 1.47, 0.80, 1.58, 1.53, 4.33, 4.23, 3.25, 3.22)
group_ind = c(rep("good", 8),
               rep("poor", 8))

obs_diff = mean(obs[group_ind == "good"]) - mean(obs[group_ind == "poor"])
obs_diff
```

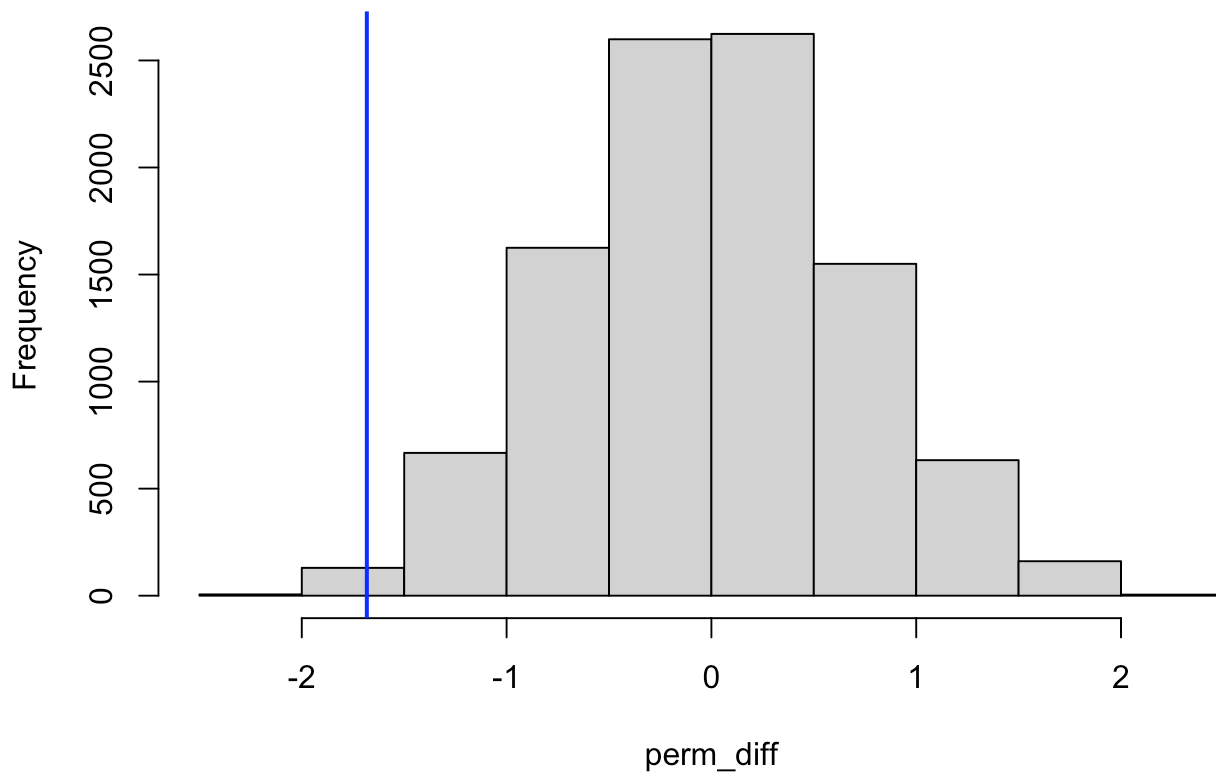
```
## [1] -1.6825
```

```
set.seed(123)
sample(group_ind)
```

```
## [1] "poor" "poor" "good" "poor" "poor" "good" "good" "good" "good" "poor"
## [11] "good" "good" "poor" "poor" "good" "poor"
```

```
permutation = function(){  
  temp_group_int = sample(group_ind)  
  diff = mean(obs[temp_group_int == "good"]) - mean(obs[temp_group_int == "poor"])  
  return(diff)  
}  
  
perm_diff = replicate(10000, permutation())  
hist(perm_diff)  
  
abline(v=obs_diff, lwd= 2, col = "blue")
```

Histogram of perm_diff



```
mean(abs(perm_diff) > abs(obs_diff))
```

```
## [1] 0.015
```